JIAHAO XIA

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EDUCATION

University of Maryland, College Park, MD — Aug 2025 – May 2028 (expected)
B.S. in Computer Science, Minor in Mathematics — GPA: 3.67/4.00

Pennsylvania State University, University Park, PA — Aug 2024 – May 2025
Electrical Engineering coursework (transferred) — GPA: 3.94/4.00

TECHNICAL SKILLS

- **Languages:** C/C++, Python, CUDA, Java, OCaml, Bash
- **ML & GPU Systems:** PyTorch, Triton, vLLM, GPU programming
- **Compilers & PL:** Polyhedral compilation, Datalog/Soufflé, Rocq/Coq
- **Systems & Tools:** Linux, Git, Docker, CMake, GDB

RESEARCH EXPERIENCE

Undergraduate Researcher — 2025 – Present
Advised by Prof. Jingbo Wang, Purdue University

PSoufflé: Scalable Probabilistic Logic Inference [GitHub]

- Extended a C++ Datalog compiler and runtime to support exact probabilistic inference for taint, side-channel, and disassembly analyses.
- Designed an independence-aware optimization achieving a **119× average end-to-end speedup** over the state of the art, and solving instances on which prior systems timed out.
- Owned the evaluation end to end: benchmark design, experiment automation, performance analysis, and a publicly released reproducible artifact.

End-to-End Validation for Polyhedral Compiler Optimizations

- Built an end-to-end compiler validation pipeline covering affine scheduling, tiling, loop reconstruction, and parallel annotations.
- Integrated PolCert's verified transformation validators to detect and localize **5 confirmed Pluto miscompilations**, including faults in tiling, scheduling, and parallelization.

OPEN SOURCE & SELECTED PROJECTS

Helion: GPU Kernel Compiler — Python, Triton [PR #3612: Under Review]

- Implemented two-way `torch.chunk` and size-two `torch.unbind` lowering for Helion's Triton backend, including Tensor method dispatch and compile-time shape validation.
- Fixed `torch.stack` code generation for flattened tiles; added regression tests for axis mapping, loop-carried accumulators, and autotuning.

Needle: Deep Learning Systems Framework — C++, CUDA, NumPy, Python [GitHub]

- Built a PyTorch-like framework from scratch with reverse-mode automatic differentiation, tensor operators, neural-network modules, and optimizers.
- Implemented native C++/CUDA tensor backends and GPU kernels with device-agnostic execution across NumPy, CPU, and CUDA.
- Trained CNNs, RNNs/LSTMs, and Transformers end to end on the resulting stack.

PUBLICATIONS

- Xuyang Li*, **Jiahao Xia***, Ahmed Adnan, Jingbo Wang. "PSOUFFLÉ: Scaling Exact Probabilistic Logic Inference for Program Analysis." *FMCAD 2026*. [PDF] [Artifact]
- Xuyang Li, Minghai Lu, Siyu Chen, **Jiahao Xia**, Weiyi Chen, Jingbo Wang. "End-to-End Verified Compilation for Polyhedral Loop Transformations." *Under review, CPP 2027*.

*Equal contribution.